CENG 454 – Artificial Intelligence and Data Science

FINAL PROJECT DETAILS

Project Objective:

The final project for CENG 454 aims to offer students the opportunity to explore advanced and creative applications in artificial intelligence and data science, with a strong focus on large language models (LLMs), generative AI, and real-world data-driven problems. Students are expected to formulate a concrete research question, implement and evaluate a method, and deliver a scientific report.

This project is intended to develop:

- A practical understanding of LLMs, ML algorithms, and AI tools,
- The ability to conduct data-driven experimentation and perform critical evaluation,
- Proficiency in scientific writing, reproducibility practices, and research ethics.

Submission Details:

- The project must be completed in groups of 3 or 4 students.
  - A list of groups is attached to this file. No changes will be made to the groups under any circumstances, regardless of the reason.
- Plagiarism is strictly prohibited. All resources utilised must be clearly cited in the bibliography section using IEEE format.
- **Submission platform:** UBYS
- **Deadline:** 22 May 2026, Sunday - 23:59
- The project should be submitted as a single `.zip` file by one member of the group.
- **Required Submission Package:** a `.zip` file should include the followings:
  - Final `.pdf` report
  - Source code (`.ipynb`, `.py`, etc.)
  - Dataset(s) or link(s) to datadet(s)
- **Submission Package Naming Format:** `CENG454_Group_GroupNo.zip`
  - *example:* `CENG454_Group_0.zip`

Project Report Details:

- **Submission Format:** `.pdf`
- **Report Length:** Must not exceed 8 pages (excluding the cover page and bibliography).
- **Cover Page Must Include:** University/Faculty/Department name, course code and name, student number and name, submission date.
- **Formatting:**
  - Font: Times New Roman
  - Font size: 11 pt
  - Line spacing: 1.0
  - One line spacing between paragraphs
  - Page margins: 2.5 cm on all sides
- Students are encouraged to include figures or tables to support their analysis. Each figure/table must be numbered and titled (e.g., “Figure 1: Quantum vs Classical Memory Hierarchy”)

Project report Structure (Recommended):

1. ***Introduction and Motivation***
   Present your research problem, motivation, and real-world relevance.
2. ***Background and Related Work***
   Discuss prior work, key papers (at least 10 relevant and recent sources), and briefly explain the concepts relevant to your work.
3. ***Dataset and Preprocessing***
   Describe the dataset source, structure, size, preprocessing steps, and rationale.
4. ***Proposed Approach / Implementation***
   Explain your experimental setup, selected algorithm(s), any novelty (e.g., prompt engineering, hybrid model design, experiments).
5. ***Evaluation and Results***
   Present metrics (e.g., accuracy, F1, BLEU, perplexity, NDCG), comparisons, and figures/tables for performance analysis.
6. ***Discussion and Limitations***
   Interpret the results, draw insights, discuss what worked and what did not.
7. ***Conclusion and Future Work***
   Summarize findings and propose ideas for improving or extending the project.
8. ***References***
   Cite all sources in IEEE citation style.

Demo Presentation:

- Groups will present their project for a brief ~10-15 minutes demo. (Further details regarding the schedule and location will be announced later.)
- All members of the group are responsible for the entire project.
- The demo is mandatory, and all group members must attend. Projects without a demo will not be evaluated, and members who do not attend will be considered as not having submitted the assignment.

Suggested Topics:

Students may choose one of the following themes or propose an original topic:

- Evaluating hallucinations in responses from ChatGPT, Claude, or Gemini
- Prompt engineering strategies for domain-specific question answering
- Bias detection in LLMs for multilingual inputs (e.g., Turkish vs. English)
- Ethical dilemmas in AI-generated legal or medical advice
- Instruction tuning vs. reinforcement learning from human feedback (RLHF)
- Fake news detection using transformer-based sentence embeddings
- Zero-shot and few-shot performance of LLMs on Turkish datasets
- Conversational agents with memory: simulation and real deployment
- Benchmarking vector search in retrieval-augmented generation (RAG)
- Sentiment shift in LLM responses under adversarial prompts
- Visual question answering using BLIP/CLIP+LLM combinations
- Language model fine-tuning for academic writing assistance
- Music or poetry generation using LLMs and style transfer techniques
- Code generation evaluation using LLMs vs. traditional IDE-based coding
- Human-AI collaboration in creative writing: experimental evaluation

Open Dataset Repositories:

- Hugging Face Datasets: https://huggingface.co/datasets
- Kaggle Datasets: https://www.kaggle.com/datasets
- Google Dataset Search: https://datasetsearch.research.google.com
- Papers With Code: https://paperswithcode.com/datasets
- TÜİK (for Turkish social/economic data): https://data.tuik.gov.tr

Evaluation Criteria

- *Originality and Relevance (25%)*
  - The students are expected to formulate a concrete and original research question that addresses real-world relevance in AI or Data Science, moving beyond standard examples.
- *Scientific Soundness and Evaluation (25%)*
  - The project must demonstrate a practical understanding of AI/ML algorithms through a methodology grounded in prior work (at least 10 sources), supported by rigorous data-driven experimentation and critical evaluation using appropriate metrics.
- *Clarity and Depth of Analysis (20%)*
  - The report should provide a proper interpretation of the results, drawing deep insights into both successes and failures, while explicitly documenting.
- *Reproducibility and Code Quality (15%)*
  - Students must ensure the reproducibility of their work by submitting a complete package, including clean source code, detailed dataset descriptions, and preprocessing rationales.
- *Presentation, Structure, and Writing (15%)*

Academic Integrity

All work must be done by the students and must cite relevant sources when using external ideas or code. Any use of AI tools (ChatGPT, Copilot, Gemini etc.) must be explicitly documented. Plagiarism or excessive reliance on generative tools without critical analysis will be penalised.

GroupNo	StudentNo	StudentFullName
1	210401012	Ali Dabanca
	210401062	Egemen Yaşamış
	210401030	Nazlı Tokgöz
	210401020	Buse Berfin Halefoğlu
2	210401002	Batuhan Bilecen
	210401018	Mustafa Onur Bayram
	210401052	Batuhan Türkaslan
	210401029	Efekan Kandak
3	210401026	Cem Barış Özden
	220401034	Özlen Yudum Besleme
	210401046	Utku Gökdağ
4	210401048	Sevcihan Suzan Bağcı
	210401047	Sinem Demir
	210401059	İrem Kurutepe
	210401009	Doğa Oytaç
5	220401060	Efekaan ketenci
	210401021	Alperen Balım
	220401096	Serdar Can
	210401050	Berke Şahin
6	230401114	Sarp Sünbül
	220401050	Alkım Gönenç Efe
	220401067	Damla Parlakyıldız
7	210401060	Murat Köktopçu
	210401022	Sude Dinçer
	210401015	Neslihan Mustak
8	220401080	Atilla Uğur
	220401056	Baturalp Kahveci
	220401048	Tuğba Dur
9	240401031	Anıl Can Eren
	210401037	Damla Çakır
	210401014	Halim Ağdemir
10	220401038	Tuğba Cebeci
	210401007	Göktuğ Ormanlı
	210401032	Berke Benli
	210401038	Elif Alınca
11	210401033	Burcu Bulut
	210401054	Sena Yüksel
	210401011	Elif Demirtaş
	210401008	Sılagül Erol
12	210401040	Beyza Aydoğmuş
	210401055	Saimenur Özen
	220401035	Neziha Beyazıt
	210401019	Erdem Altuğ Malkan
13	220401082	Samet Ünsal
	230401034	Fatmanur Bulut
	210401035	Eren Makraç
14	210401027	Betül Sena Dalgıç
	220401040	Doğa Çalkap
	210401041	Emine İrem Demiryürek
	210401039	Esma Fatime Çavuş
15	210401057	Züleyha Demir
	220401075	Beyza Tatlıdil
	230401027	Tuana Atalay
	220401094	Cemal Özcan
16	210401031	Duygu Elif Yavuz
	230401040	Ömer Selahattin Demirci
	230401065	Göknur Kaya
	220401042	Beren Çetin
*17	220401063	Ahmet Selim Koç
	220401065	Necati Kurtaran
	210401053	Ezgi Mesut
	220401049	Süleyman Alp Durmaz

* Groups formed randomly from students who did not join any group.